The German University in Cairo
Electronics Dept., Faculty of IET
Semester: 6th Electronics
Course: Electronic Circuits (ELCT 604)
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Audio Amplifier Design and Implementation

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1. Introduction & Design Modifications

Introduction

The goal of this project is to design, simulate, and implement a BJT-based audio amplifier. The circuit is designed to take a small-signal audio input and provide sufficient voltage and current gain to drive a standard low-impedance load (4-ohm or 8-ohm speaker). The project includes theoretical hand analysis, simulation, and a hardware breadboard implementation.

Design Modifications

The original schematic provided in the project description proposed using BC337 transistors for the final output stage. However, BC337 transistors have a relatively low maximum collector current (I_C) and power dissipation rating. Driving an 8-ohm speaker directly with them would result in severe signal clipping and potential thermal failure.

To safely and effectively drive a standard speaker, the output stage was redesigned:

- **Class AB Power Stage:** We replaced the BC337 output stage with a push-pull Class AB amplifier using TIP35 (NPN) and TIP36 (PNP) power transistors. These devices are designed to handle the high current demands of a low-impedance load without clipping.
- **V_{BE} Multiplier:** To drive the TIP transistors, a V_{BE} multiplier circuit (Q1) was added. This provides a constant DC voltage drop to bias the TIP35 and TIP36 slightly into conduction, which eliminates the crossover distortion that would otherwise occur when the AC signal crosses zero.

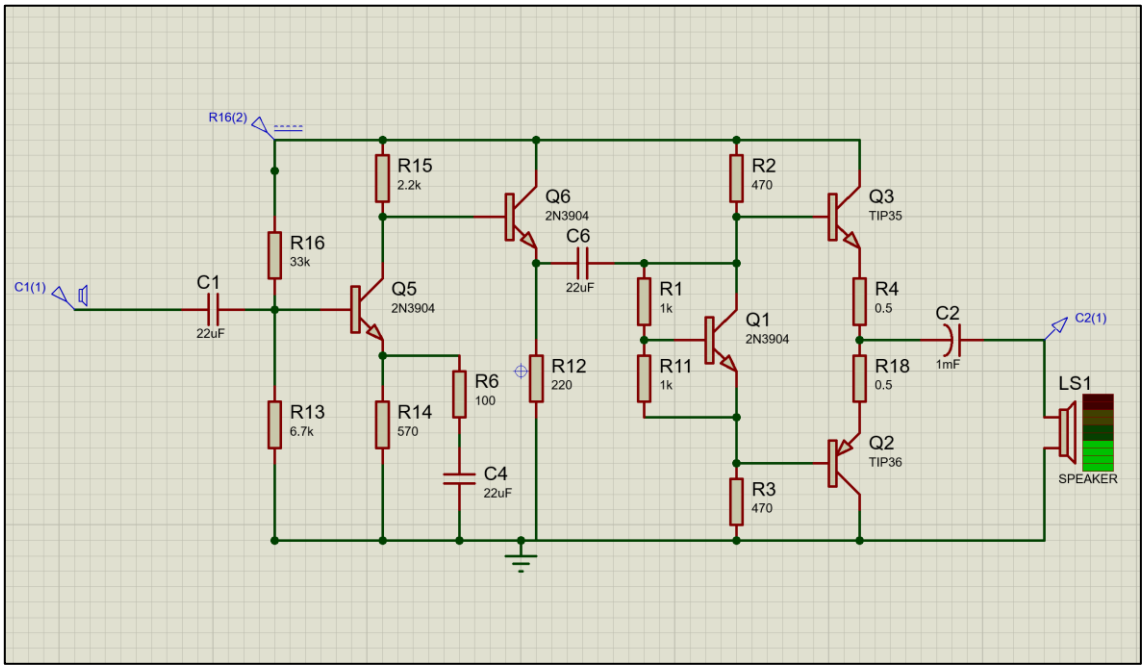


Figure 1: Complete schematic of the modified Class AB audio amplifier.

2. Circuit Architecture & Block Diagram

The amplifier is divided into four distinct functional stages to process the small-signal audio input and deliver a higher current to the 4-ohm or 8-ohm speaker.

Stage	Core Components	Function / Description
Input & Voltage Gain	C1, Q5, R13, R16, R15, R14, R6, C4	C1 couples the AC input. Q5 (2N3904) operates as a Common-Emitter amplifier providing the primary voltage gain. C4 partially bypasses the emitter resistance to boost AC gain while maintaining DC stability.
Buffer Stage	Q6, R12, C6	Q6 operates as an Emitter-Follower (Common-Collector) buffer. It prevents the next stage from loading down the high-impedance collector of Q5, translating the signal to a lower output impedance.
Biasing Network	Q1, R1, R11	Acts as a V_{BE} multiplier. It maintains a constant DC voltage drop between the bases of the power transistors to bias them slightly above cut-off, eliminating crossover distortion.
Class AB Output	Q3 (TIP35), Q2 (TIP36), R4, R18, C2	A push-pull current amplifier. Q3 handles the positive half-cycle and Q2 handles the negative half-cycle, providing the necessary current to drive the low-impedance speaker (LS1). C2 blocks DC from entering the speaker.

3. DC Hand Analysis (Quiescent Point)

The following theoretical hand calculations were performed to establish and verify the quiescent operating points (Q-points) for the initial amplification stages. Establishing a stable DC baseline is critical to ensure the transistors remain in the forward-active region, thereby maximizing the available voltage swing for the AC audio signal and preventing premature clipping.

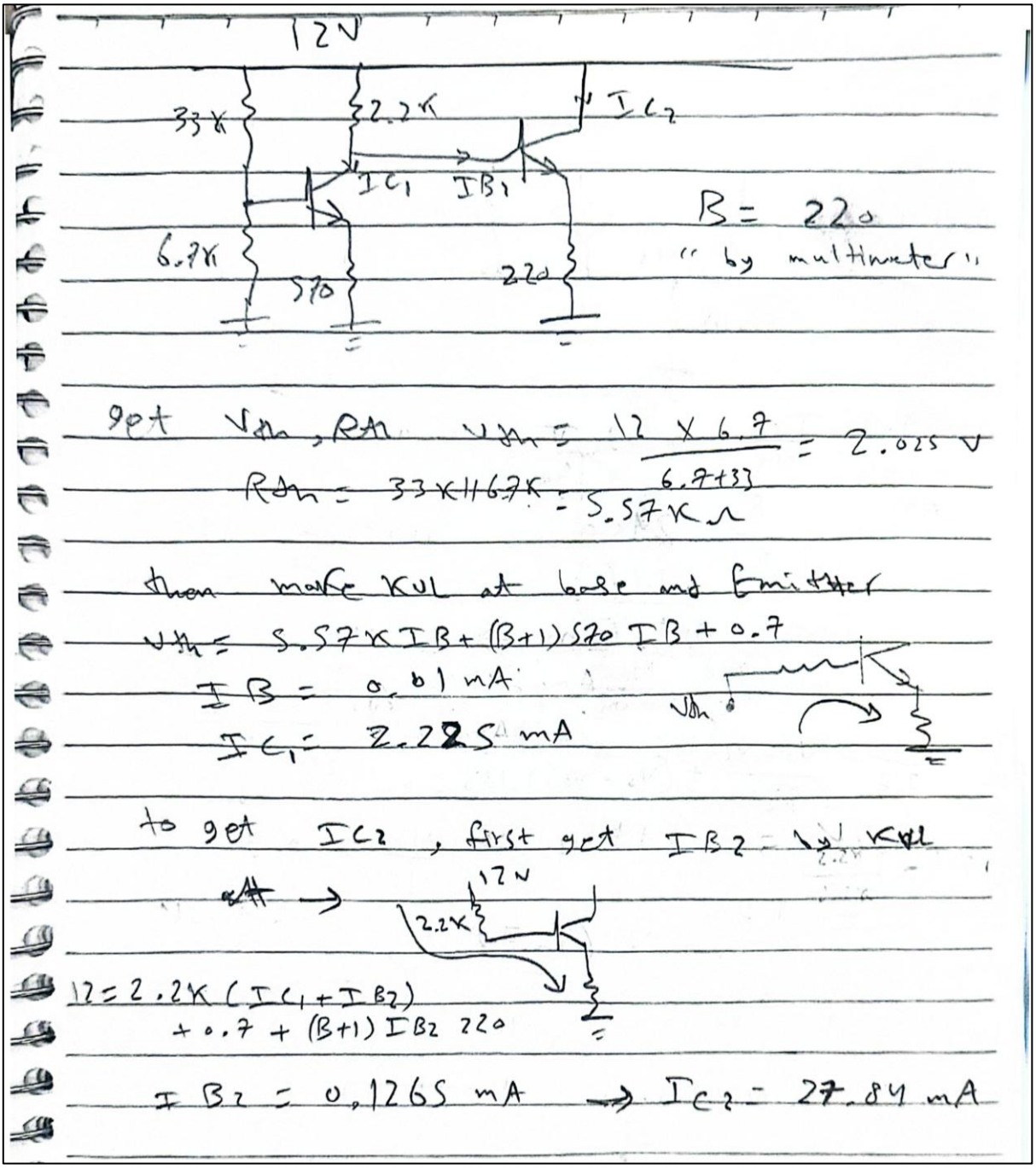


Figure 2: Hand analysis of DC quiescent operating points.

Key Analytical Takeaways:

- **Thevenin Equivalent Bias:** Analyzing the input voltage divider network yields a base Thevenin voltage (V_{th}) of $2.025V$ and a Thevenin resistance (R_{th}) of $5.57\ k\Omega$.
- **Measured Transistor Parameters:** To ensure the theoretical calculations closely mirror the physical hardware behavior, a precise, multimeter-measured β value of 220 was utilized for the BJT calculations rather than assuming a generic datasheet value.
- **Quiescent Collector Currents:** Applying Kirchhoff's Voltage Law (KVL) to the base-emitter loops confirms that the primary Common-Emitter stage operates with a collector current (I_{C1}) of $2.225\ mA$. The subsequent stage operates at a higher quiescent current (I_{C2}) of $27.84\ mA$, ensuring it is properly biased to drive the following buffer and push-pull output stages.

4. AC Hand Analysis (Small Signal Model)

The following theoretical hand calculations utilize the r_π small-signal equivalent circuit (hybrid-pi model) to determine the unloaded input and output impedances, as well as the mid-band voltage gain of the pre-amplifier stages.

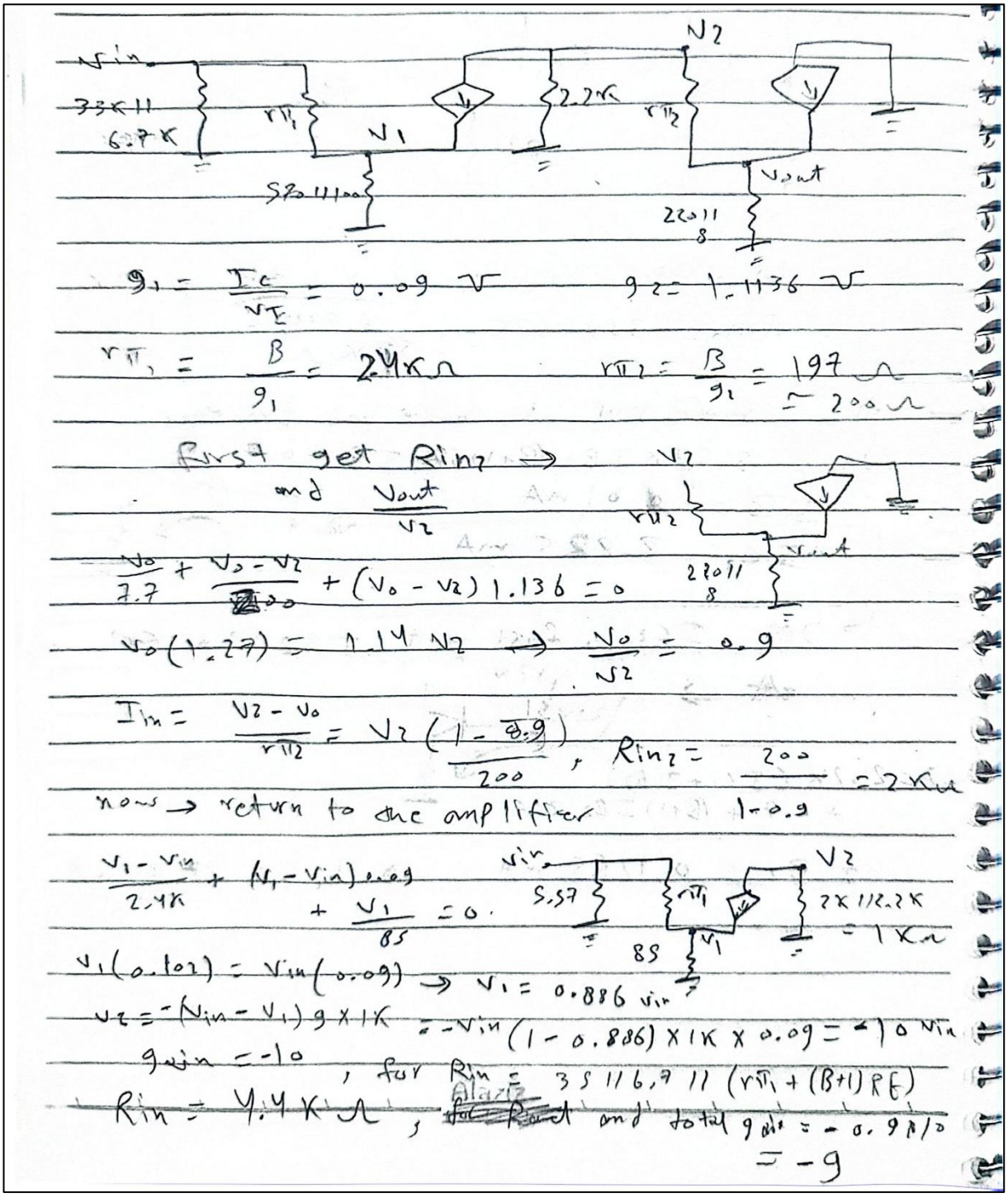


Figure 3: Small-signal AC hand analysis, covering Hybrid-Pi parameter extraction, initial input resistance (R_{in}), and stage gain derivations.

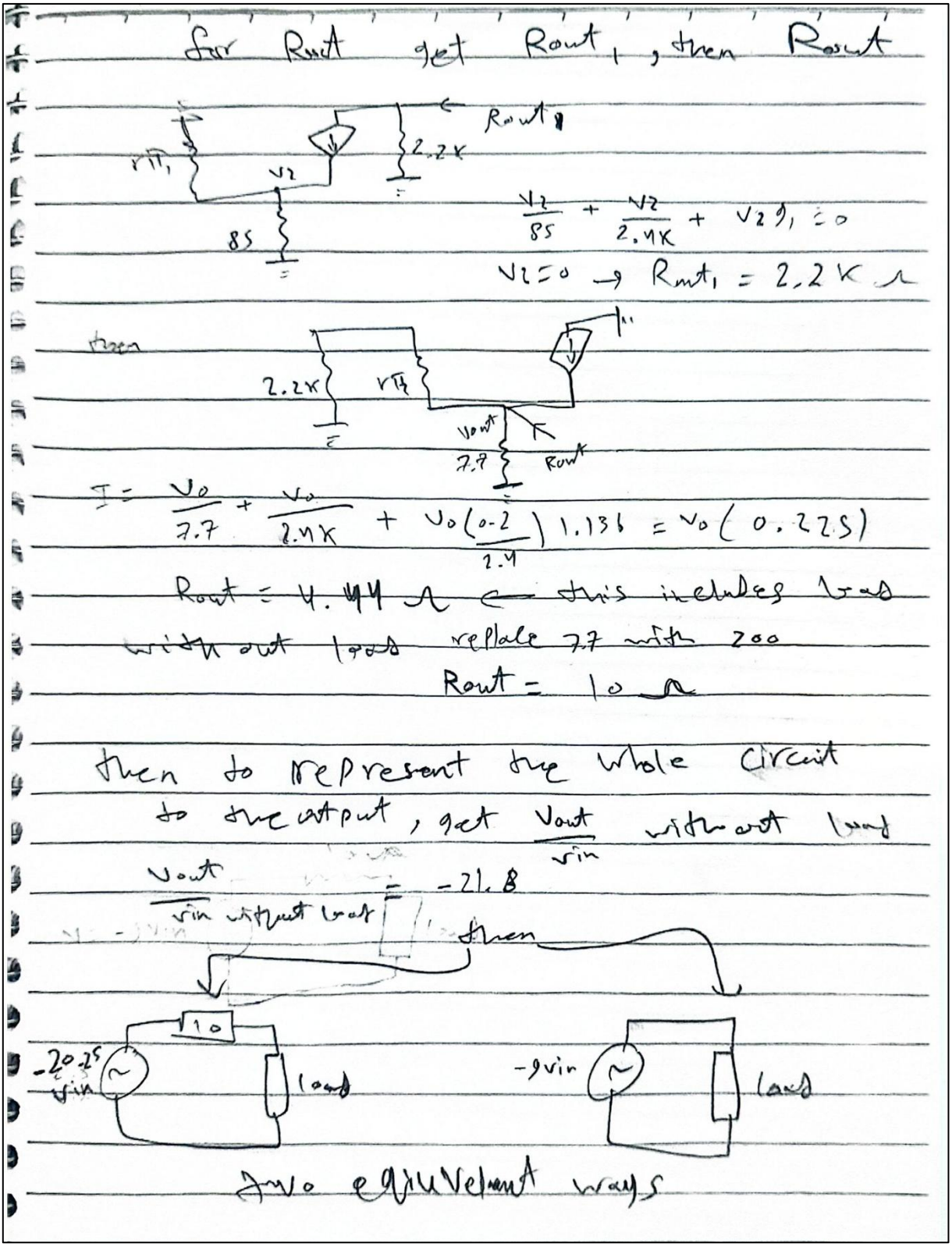


Figure 4: Continued small-signal AC hand analysis, detailing the pre-amplifier's output impedance (R_{out}) derivation and unloaded versus loaded gain models.

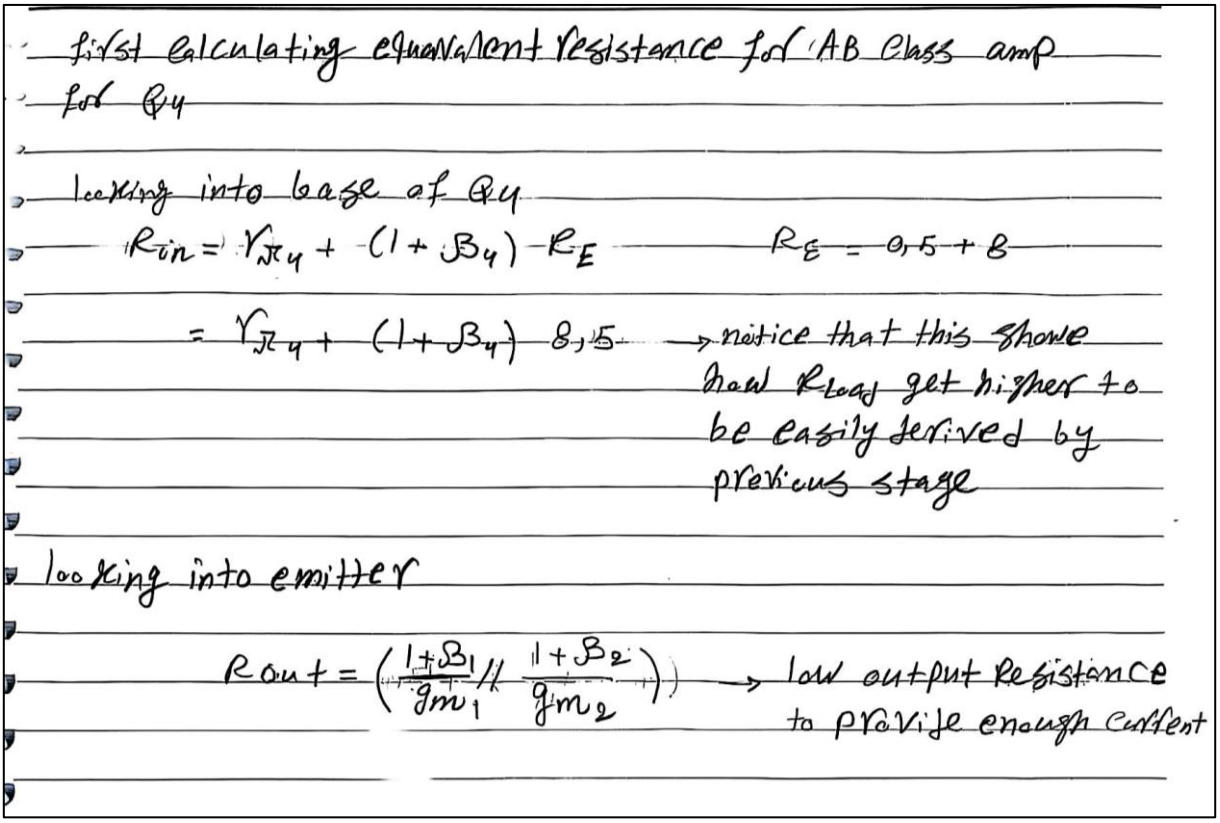


Figure 5: Theoretical derivation of the Class AB output stage input resistance (R_{in}) and parallel emitter output impedance (R_{out}).

Key Analytical Takeaways:

- **Hybrid-Pi Parameter Extraction:** Utilizing the DC quiescent currents established in the biasing section, the dynamic small-signal parameters (specifically transconductance g_m and base input resistance r_π) were calculated for the active devices. These derived values formed the mathematical foundation of the AC equivalent circuit.

- **Theoretical Gain:** The hand analysis maps the equivalent circuit to calculate an absolute maximum theoretical voltage gain of $-21.8\ V/V$ for the isolated Common-Emitter stage. Furthermore, the mathematical model correctly accounts for cascaded loading; it predicts that the Emitter-Follower buffer's finite input resistance (calculated at $2\ k\Omega$) will load the primary collector, attenuating the effective theoretical pre-amplifier gain to $-10\ V/V$ before the final power stage is attached.

- **Thevenin Output Impedance & Stage Justification:** The small-signal equivalent circuit successfully established the pre-amplifier's internal driving impedance, yielding a theoretical output resistance (R_{out}) of exactly 10 ohms at the buffer's emitter. Identifying this baseline is a critical analytical step, as it provides the mathematical justification for the cascaded Class AB power stage. By acting as an impedance buffer, the Class AB stage presents a sufficiently high input resistance to prevent choking the pre-amplifier's gain, while simultaneously stepping down the final output resistance to a fraction of an ohm, ensuring efficient power transfer to the heavy 8-ohm load.

- **Class AB Impedance Transformation:** The analysis of the push-pull stage mathematically proves its function as a critical impedance buffer. The input resistance derivation ($R_{in} = r_\pi + (1+\beta)R_E$) demonstrates that the heavy 8 Ω load is multiplied by the transistor's current gain (β). This presents a sufficiently high impedance to the pre-amplifier, preventing severe voltage attenuation. Conversely, the output resistance (R_{out}), formed by the parallel combination of the internal dynamic emitter resistances $\left(\frac{1}{g_{m1}} \parallel \frac{1}{g_{m2}}\right)$, results in an ultra-low theoretical driving impedance, ensuring maximum and efficient current delivery to the speaker.

5. Simulation Results (KiCad & Proteus)

5.1 DC Operating Point Verification:

To validate the theoretical biasing network before dynamic AC testing, a DC operating point (.op) simulation was executed. The analysis confirms that the quiescent parameters of all cascaded stages meet the precise requirements for a single-supply audio amplifier.

- **Voltage Gain & Isolation Stages:** The primary Common-Emitter stage is biased centrally in its active region, drawing 1.95mA with a V_{CE} of $5.97V$ to prevent premature signal clipping. The subsequent Emitter-Follower buffer effectively tracks this DC level, outputting $6.22V$ while drawing $28.0mA$ to provide necessary current drive for the output stage.

- **Biasing & Power Stages:** The adjustable V_{BE} multiplier performs perfectly, generating a $1.86V$ potential difference that biases the TIP41C and TIP42C complementary pair slightly above cut-off. This is evidenced by the continuous $33.1mA$ quiescent current flowing through the power stage, effectively mitigating crossover distortion. Furthermore, the push-pull emitter junction stabilizes exactly at 6.00V. Centering the output at precisely half of the 12V supply rail is critical, as it allows for the absolute maximum symmetrical voltage swing before rail-induced clipping occurs.

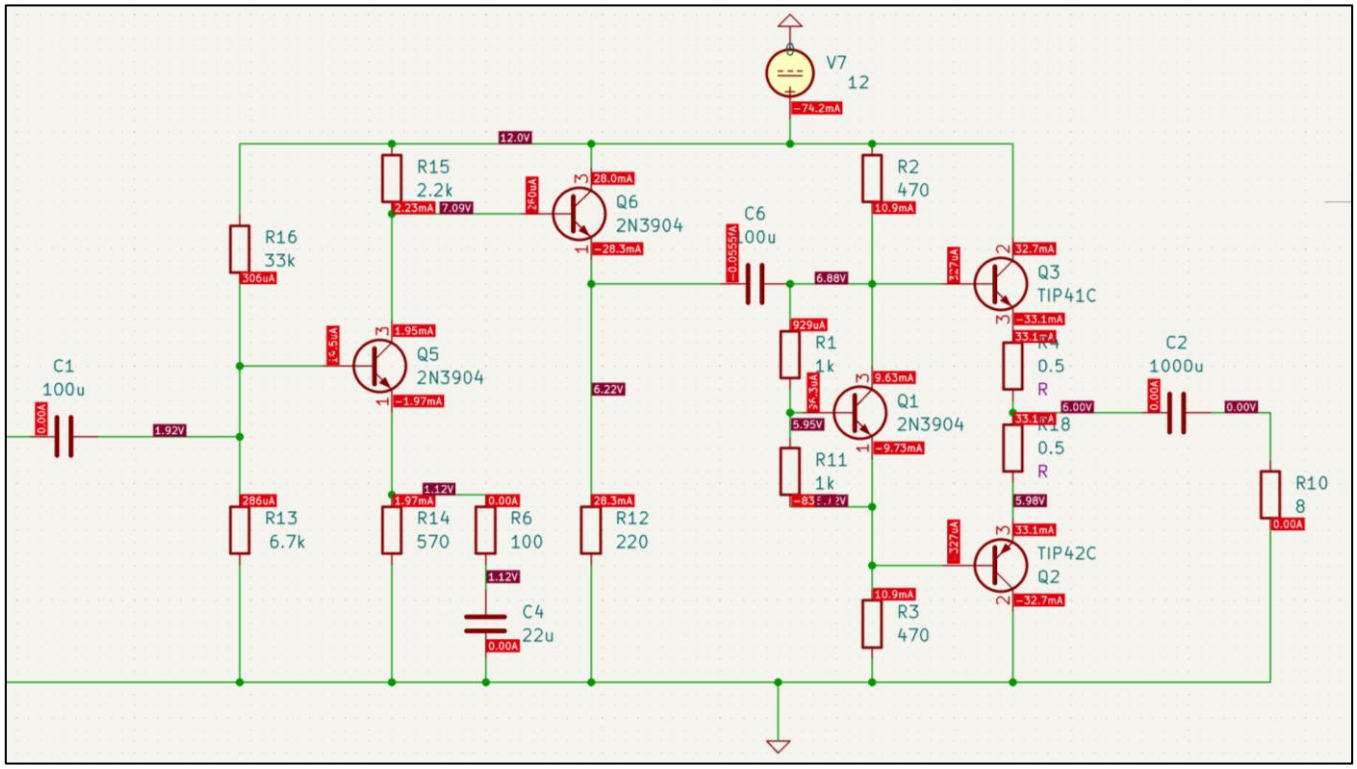


Figure 6: KiCad DC Operating Point verification showing 6.00V output centering.

5.2 AC Transient Analysis: Unloaded Pre-Amplifier

To isolate and verify the primary voltage gain stage against the theoretical small-signal model, an unloaded AC transient simulation was performed. The Class AB output stage was temporarily decoupled, and the pre-amplifier buffer was AC-coupled into a 1 MΩ dummy load. This extreme resistance effectively simulates an open-circuit state, preventing SPICE floating-node errors while ensuring zero current loading on the buffer.

A 1 kHz, 200mV peak-to-peak sine wave was injected at the input coupling capacitor. The simulation yielded an inverted output swing of 4.05V peak-to-peak across the open-circuit load. This results in a realized, unloaded mid-band voltage gain of −20.26 V/V.

This simulated value acts as a highly accurate validation of the theoretical hand calculations. The theoretical model predicted a maximum unloaded gain of -21.8 V/V. The minor attenuation observed in the SPICE environment (from 21.8 down to 20.26) physically represents the real-world loading effect of the internal Emitter-Follower buffer, proving the system is functioning optimally before the heavy 8-ohm speaker load is introduced.

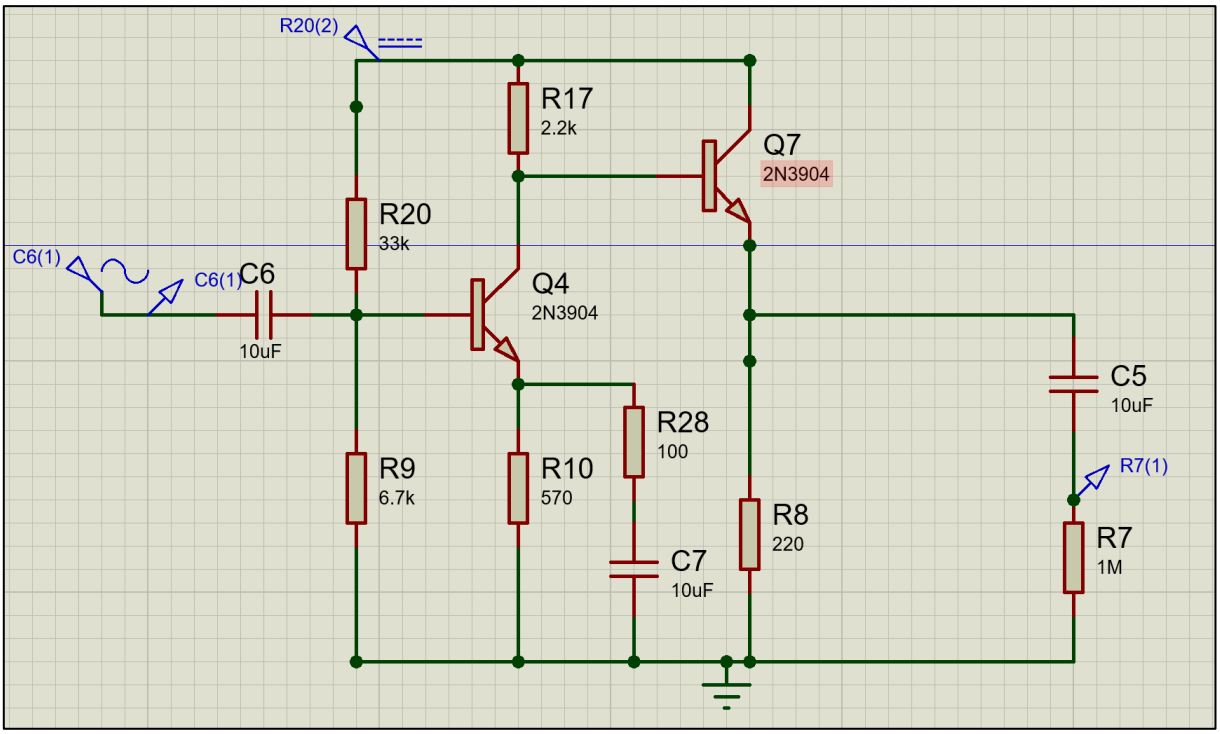


Figure 7: Proteus Pre-Amplifier Circuit Schematic.

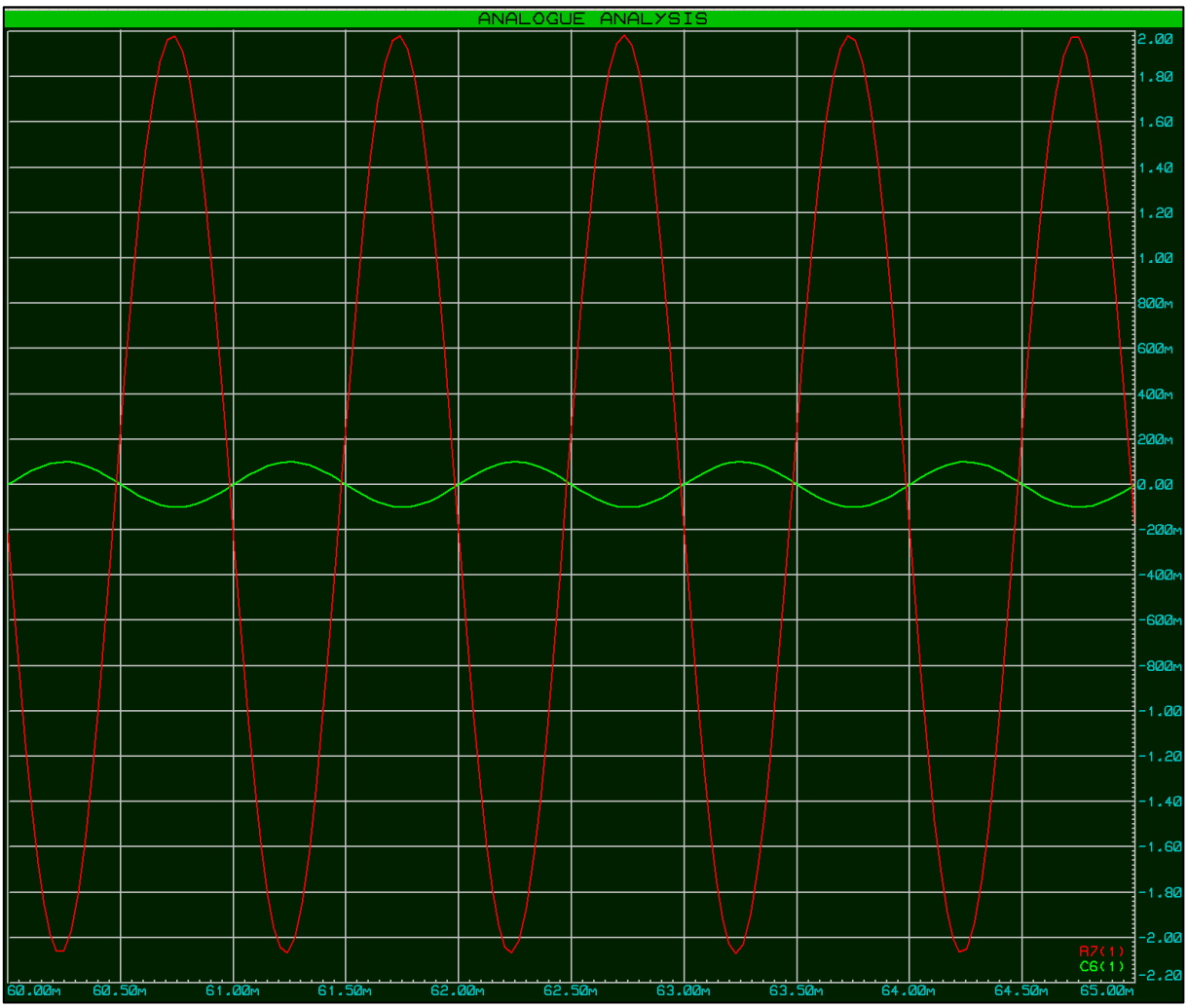


Figure 8: Proteus transient analysis of the unloaded pre-amplifier (-20.26 V/V gain).

5.3 AC Transient Analysis: loaded Pre-Amplifier (Thevenin Verification)

To physically verify the small-signal equivalent models derived in the theoretical hand analysis, an intermediate loading test was simulated. The Class AB push-pull stage remained disconnected, and a 10 Ω resistive dummy load was attached directly to the Emitter-Follower buffer output via a 1 mF coupling capacitor. The coupling capacitor was upsized to 1 mF to ensure a negligible reactive impedance ($X_c \approx 0$) at the 1 kHz testing frequency, thereby preventing unwanted AC phase shifts. The input signal was scaled down to 20mV peak-to-peak to prevent severe clipping at the buffer stage.

The simulation yielded an output swing of approximately 160mV peak-to-peak across the 10 Ω load, resulting in a loaded voltage gain of $\mathbf{-7.98\ V/V}$.

This severe attenuation from the unloaded gain ($-20.26\ V/V$) physically demonstrates the voltage divider effect caused by the pre-amplifier's internal Thevenin output impedance. By analyzing the voltage drop ($V_{out} = V_{internal} \times [R_{load}/(R_{load} + R_{out})]$), the simulated internal output resistance (R_{out}) of the buffer stage was mathematically extracted to be $\mathbf{15.4\ \Omega}$. This closely corroborates the idealized hand calculation of 10 Ω. The minor discrepancy highlights the accuracy of the SPICE engine, which dynamically accounts for current-dependent β reduction and internal emitter resistance (r_e) that static theoretical models often omit. Ultimately, this 15.4 Ω simulated baseline mathematically proves the absolute necessity of adding the subsequent Class AB stage to properly drive an 8 Ω speaker.

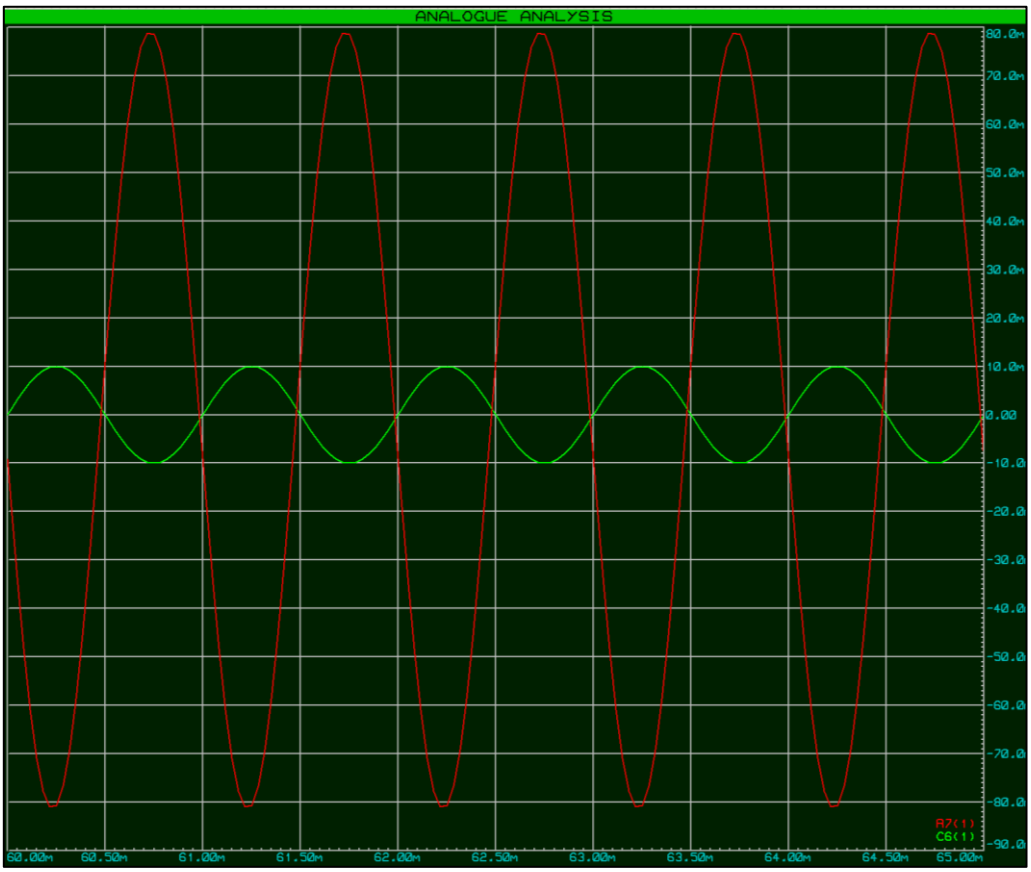


Figure 9: Proteus transient analysis of the 10 Ω load test (-7.98 V/V gain).

5.4 AC Transient Analysis: Fully Loaded Class AB System

The final phase of simulation evaluated the complete, fully loaded cascaded architecture using Proteus. The Class AB power stage composed of TIP35C and TIP36C complementary transistors were integrated with the pre-amplifier, and the system was loaded with an 8 Ω speaker model via a 1 mF coupling capacitor.

Stimulating the primary input with a 1 kHz, 20mV peak-to-peak sine wave produced an undistorted, inverted output of 340mV peak-to-peak across the speaker, corresponding to a definitive system voltage gain of -17.0 V/V.

Impedance Transformation Analysis:

This loaded gain figure practically demonstrates the mechanical necessity of the push-pull topology. As proven in the 10 Ω dummy load test, the pre-amplifier possesses an inherent 15.4 Ω output resistance; coupling this directly to an 8 Ω load results in a severe voltage divider effect that collapses the gain.

The insertion of the TIP35C/TIP36C stage rectifies this mismatch. Operating as high-current Emitter-Followers, the power transistors divide the pre-amplifier's output resistance by their respective β values, transforming the system's final Thevenin driving impedance to a fraction of an ohm. This impedance shielding prevents excessive voltage attenuation and allows the pre-amplifier's voltage swing to be efficiently transferred as high-current driving power to the 8 Ω load, all while maintaining a continuous waveform free of crossover distortion.

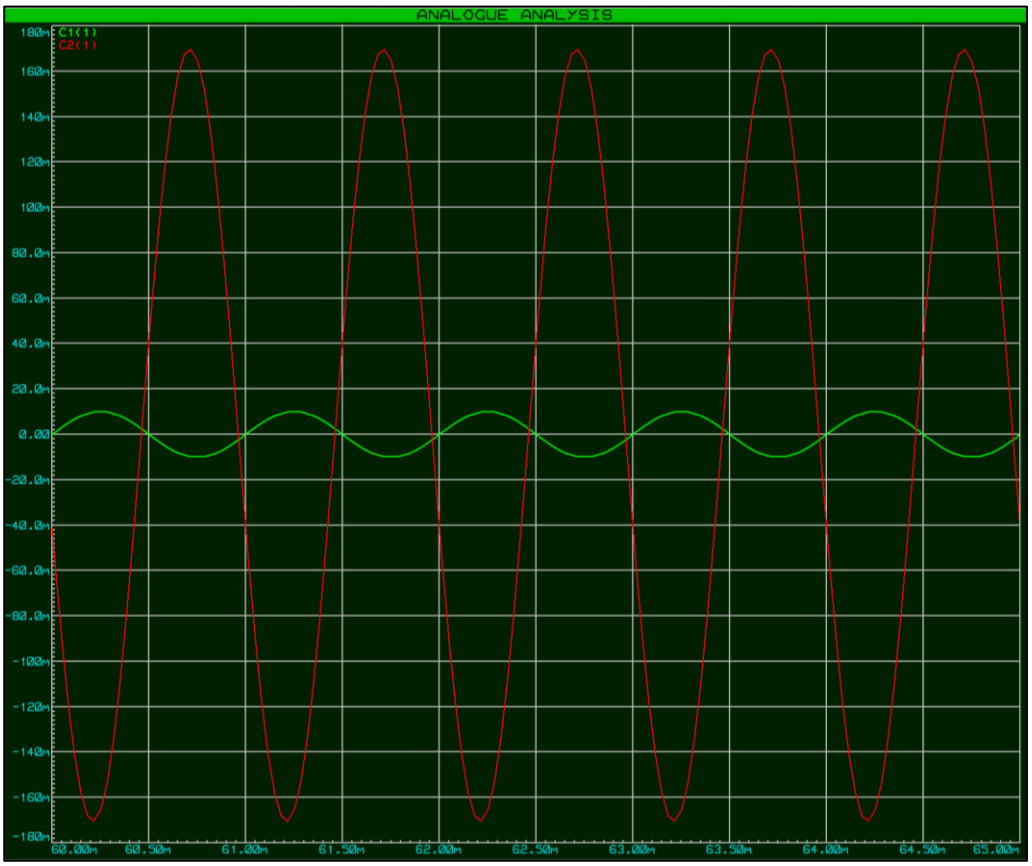


Figure 10: Proteus transient analysis of the fully loaded Class AB system (-17.0 V/V gain).

6. Hardware Implementation & Physical Verification

6.1 Circuit Construction & Input Biasing

The cascaded amplifier was physically constructed on a standard breadboard to validate the transition from SPICE models to a functional system. The primary voltage gain stage and isolation buffer (2N3904) were integrated with a high-current Class AB output stage (TIP35C/TIP36C). A condenser microphone was used as the primary input source, supported by a DC biasing network to provide necessary operating voltage while maintaining AC signal integrity via an input coupling capacitor.

6.2 Real-World Measurement & Results

Hardware testing was conducted across all stages to verify the integrity of the design. Using a laboratory oscilloscope and function generator, the following results were obtained:

- **Pre-Amplifier Stage Verification:** Under open-circuit conditions, the first stage produced a real-world voltage gain of approximately -20 V/V, showing excellent alignment with the theoretical small-signal model and the Proteus simulation.
- **Full System Verification:** The final Class AB output stage was integrated and tested with a physical 0.5W, 8 Ω loudspeaker. The hardware yielded results highly consistent with the Proteus simulation, maintaining the projected system gain of approximately -17.0 V/V.
- **Acoustic Integrity:** The system successfully drove the loudspeaker with a clear, audible signal from the condenser microphone. The V_{BE} multiplier successfully eliminated crossover distortion in the physical implementation, matching the smooth transitions observed in the simulation waveforms.

6.3 Error Margin and Discrepancy Analysis

The hardware results indicate that the physical circuit operates within a narrow margin of error compared to the software models. The minor variances observed are attributed to standard engineering factors:

- **Component Tolerance:** The 1-5% tolerance of resistors and the inherent variance in BJT forward current gain (β) across discrete units.
- **Parasitic Impedances:** Breadboard trace inductance and contact resistance, which can introduce minor attenuation.
- **Supply Stability:** Minor fluctuations in the 12V DC rail during high-current output peaks.

7. Conclusion

This project successfully demonstrates the design and implementation of a discrete BJT audio amplifier. By systematically addressing the impedance mismatch between high-gain voltage stages and low-impedance acoustic loads, a functional system was developed that provides significant power gain without signal distortion. The close correlation between the hand calculations, Proteus/KiCad simulations, and real-world hardware measurements specifically the consistent -17 V/V gain under load validates the efficacy of the cascaded architecture. The final system stands as a proven platform for low-power audio applications, with the potential for future expansion into Bluetooth-integrated receivers.